

BASICS

Probability space $(\Omega, \mathcal{F}, P)$
 Ω : sample space, all possible outcomes
 $\mathcal{F}$: all allowable events (set of outcomes)
 P : probability measure, assigns probability to events

ME: mutually Exclusive
CE: collectively Exhaustive

Kolmogorov's Axioms
 1. $P(A) \geq 0$ (non-negative probabilities)
 2. $P(\Omega) = 1$ (probability of nothing happening is 0, all of Ω)
 3. $P(A \cup B) = P(A) + P(B)$ for disjoint events $A, B, \dots$

Consequences:

$P(\emptyset) = 0$
 $P(A^c) = 1 - P(A)$
 $P(A \cap B) = P(A)P(B)$ if independent
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $P(A \cap B) = P(A)P(B)$ if independent
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If $A_1, A_2, \dots, A_n$ are ME & CE, A_i 's PARTITION Ω

$P(\Omega) = \sum_{i=1}^n P(A_i \cap \Omega) = \sum_{i=1}^n P(A_i)P(\Omega)$

Independence
 $P(A \cap B) = P(A)P(B)$
 $P(A \cap B \cap C) = P(A)P(B)P(C)$

Permutations of size k : $n! / (n-k)!$
 in n objects: $n!$

Partition n objects: $n!$
 into n groups: $n!$

DISCRETE
 $P(X=x) = P_X(x)$ prob. mass func. (PMF)
 $P_X(x) = P(X=x)$
 $P_X(x) = P(X=x)$

Distribution	Intuition	PMF	$E[X]$	$Var(X)$
Bernoulli $X \sim \text{Ber}(p)$	one coin flip w.p. p	$P(X=1) = p$ $P(X=0) = 1-p$	p	$p(1-p)$
Binomial $X \sim \text{Bin}(n, p)$	n coin flips w.p. p	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Geometric $X \sim \text{Geom}(p)$	time to first success	$P(X=k) = (1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson $X \sim \text{Pois}(\lambda)$	# events in fixed time interval	$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$	λ	λ
Uniform $X \sim \text{Unif}(a, b)$	roll a fair die	$P(X=x) = \frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2 - 1}{12}$

Misc.

Linearity of Poisson RV: $\text{Pois}(\lambda) + \text{Pois}(\mu) \sim \text{Pois}(\lambda + \mu)$

Indicator $I = \mathbb{1}_A$ if A else 0

Joint $P_{X,Y}(x,y) = P(X=x, Y=y)$

PMFs: $P_X(x) = \sum_y P_{X,Y}(x,y)$ $P_Y(y) = \sum_x P_{X,Y}(x,y)$

conditional $P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$ $P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$

PMF: $P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$ $P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$

EXPECTATION/VARIANCE (COVARIANCE)

$E[X] = \sum_x x P_X(x)$ $E[g(X)] = \sum_x g(x) P_X(x)$ $E[XY] = E[X]E[Y]$ if $X \perp Y$

$E[aX+b] = aE[X] + b$ $E[aX+bY] = aE[X] + bE[Y]$ $Var(aX+b) = a^2 Var(X)$

$Var(X) = E[(X-E[X])^2]$ $Var(aX+b) = a^2 Var(X)$ $Var(X+Y) = Var(X) + Var(Y)$ if $X \perp Y$

$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$ $Cov(X,Y) = E[(X-E[X])(Y-E[Y])]$

$Var(X) = E[X^2] - (E[X])^2$ $Var(Y) = E[Y^2] - (E[Y])^2$ $Cov(X,Y) = E[XY] - E[X]E[Y]$

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MATH STUFF

geom series

$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$ for $|r| < 1$

Taylor series

$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

Fundamental Theorems:

$\frac{d}{dx} \int_a^x f(t) dt = f(x)$

$\int_a^b f(x) dx = F(b) - F(a)$ where $F' = f$

$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$

CONCENTRATION INEQUALITIES

Markov: $IP(X \geq 0, P(X \geq a) \leq \frac{E[X]}{a})$

Chebyshev: $P(|X - E[X]| \geq c) \leq \frac{Var(X)}{c^2}$

special case: $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

Chernoff: $P(X > b) \leq \min_{t > 0} \{E[e^{tX}]\}$

For $X(0,1)$: $P(X > k) \leq e^{-k^2/2}$

CONVERGENCE

Almost Sure:

SLN: $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu$ if $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu$

Borel-Cantelli: A_n event n 's

$\sum_{n=1}^{\infty} P(A_n) < \infty \Rightarrow P(A_n \text{ infinitely often}) = 0$

$\sum_{n=1}^{\infty} P(A_n) = \infty \Rightarrow P(A_n \text{ infinitely often}) = 1$

Convergence: $P(A_n \text{ infinitely often}) = 1 \Rightarrow \text{converges}$

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POISSON PROCESSES

INTERVAL TIMES $s_1, \dots, s_n \sim \text{Exp}(\lambda)$ $T_n = \sum_{i=1}^n s_i$
 N(t) = # arrivals by time t: $N(t_1, t_2) = N(t_2) - N(t_1)$

- 1) STATIONARY INCREMENTS: $N(t, t+s) = N(s)$
- 2) INDEPENDENT: $N(t_1), N(t_2), \dots$ independent
- 3) $N(t) \sim \text{Pois}(\lambda t)$

SUM OF n i.i.d. Exp RVs: $f_{T(n)}(t) = \frac{\lambda^n e^{-\lambda t} t^{n-1}}{(n-1)!}$

$P(N(t) = n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$ Merging: $M + M \sim \text{PP}(\lambda + \mu)$
 Splitting: $M \sim \text{PP}(\lambda)$ $M \sim \text{PP}(\lambda_1 + \lambda_2)$

Memorylessness: $N(t) - N(t_0) \sim \text{Pois}(\lambda(t - t_0))$
 conditioned on # arrivals @ a time:

$$f_{s_i(n)}(s_1, \dots, s_n | n) = \frac{n!}{s_1! \dots s_n!} e^{-\lambda t} \quad E[s_i | N(t) = n] = \frac{t}{n+1}$$

$$f_{s_i(n)}(s_1, \dots, s_n | n) = \frac{n! s_i^{n-1} (t-s_i)^{n-1}}{t^{n-1} (n-1)! (n-1)!} \quad P(x_i > t | N(t) = n) = \left(\frac{t-s_i}{t}\right)^n$$

$$P(s_i > t | N(t) = n) = \left(\frac{t-s_i}{t}\right)^n \quad E[x_i | N(t) = n] = \frac{t}{n+1}$$

$$P(N_k \text{ even}) = \frac{1 + e^{-2\lambda t}}{2}$$

CONTINUOUS TIME MARKOV CHAINS (CTMC)

JUMP TIMES: spent in current state $\sim \text{Exp}(\lambda)$

JUMP TIME $\sim \text{Exp}(\lambda_1 + \lambda_2 + \dots + \lambda_n) \rightarrow$ mean time $= \frac{1}{\lambda_1 + \dots + \lambda_n}$

rate matrix: $Q(i, j) \geq 0 \quad \sum_j Q(i, j) = 0 \quad Q(i, i) = -\sum_j Q(i, j)$

$$e^{-\lambda t} = 1 - \lambda t \quad P(\min V_i = V_j) = \frac{\lambda_j}{\sum \lambda_i}$$

stationarity: $\pi Q = 0, \quad \sum \pi(i) = 1 \quad \pi(i) = \frac{1}{Z} \sum_j \pi(j, i) P(j, i)$

hitting time: $\beta(i) = \frac{1}{q_{i1}} + \sum_j \beta(j) q_{ji}$ $P^N: \mu$ of jump chain

$$\text{jump chains: CTMC} \rightarrow \text{DTMC: } \pi_i = \frac{\mu_i q_{i1}}{\sum_j \mu_j q_{j1}} \quad \mu_i = \frac{\pi_i q_{i1}}{\sum_j \pi_j q_{j1}}$$

pick $q_1 \geq \max_j q_{j1} \rightarrow P = I + \frac{1}{q_1} Q \rightarrow P(i, i) = ?$ (self loop)

birth-death: birth λ_n death $\mu_n \quad \pi_n = \pi_0 \prod_{i=1}^n \frac{\lambda_{i-1}}{\mu_i} \quad \pi_0 = \left(1 + \sum_{n=1}^{\infty} \prod_{i=1}^n \frac{\lambda_{i-1}}{\mu_i}\right)^{-1}$

$\exists \pi(i) \quad \sum_{i=0}^{\infty} \prod_{k=1}^i \frac{\lambda_{k-1}}{\mu_k} < \infty \quad P^n \Rightarrow P(\infty) = e^{\beta \beta}$

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RANDOM GRAPHS

$D = d(\text{degrees}) = \binom{2}{2} p^{d-1} (1-p)^{n-d} \quad E[D] = (n-1)p$ Binomial $(n-1, p)$

$B(n, p) \sim \text{Pois}(np)$ if $np \rightarrow \lambda \quad P(D=d) \sim e^{-\lambda} \frac{\lambda^d}{d!}$

connectivity: $P(n) = \lambda \frac{\ln n}{n} \quad \lambda < 1 \rightarrow \text{disconnected} \rightarrow 0$
 $\lambda > 1 \rightarrow \text{connected} \rightarrow 1$

ESTIMATION

MLE: $\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta) = \arg \max_{\theta} \log f(x; \theta)$

MAP: $\hat{\theta}_{MAP} = \arg \max_{\theta} L(\theta) P(\theta) = \arg \max_{\theta} (L(\theta) + \log P(\theta)) \quad P(\theta) = \frac{P(\theta) f(\theta)}{P(\theta)}$

HYPOTHESES TESTING

Type I Error: $PFA \quad P(\hat{X}(Y) = 1 | X = 0)$ (a)

Type II Error: accept null when not true $P(R(y) = 0 | X = 1)$ (b)

POD: power: correctly reject null

$\max_{\theta} P(\hat{X}(y) = 1 | X = 1) \quad \text{s.t. } P(\hat{X}(y) = 1 | X = 0) \leq \text{sig. level.}$

$\hat{X}(y) = \frac{f_{X|Y}(y|1)}{f_{X|Y}(y|0)} \quad \hat{X}(y) = \begin{cases} \text{Bayesian reason} \\ \text{Bayesian reason} \end{cases}$

$\gamma, \lambda \quad \text{s.t. } P(R(y) = 1 | X = 0) = \beta$

Find λ : use γ under H_0

$P(L(y) > \lambda | X = 0) \leq \beta \leq P(L(y) > \lambda | X = 1)$

$P(L(y) > \lambda | X = 0) = \beta \quad L \text{ continuous} \Rightarrow \text{don't worry abt } \gamma$

ODS (ODS): Find γ : $\gamma = \frac{P(L(y) > \lambda | X = 0)}{P(L(y) > \lambda | X = 1)}$

HILBERT SPACES

$X, Y \in \mathcal{H}$ $\langle X, Y \rangle = E[X \cdot Y] = E[X] \cdot E[Y]$

$\langle X, Y \rangle = 0$ orthogonal $\|X\|^2 = \langle X, X \rangle$

$\rho = \frac{|\langle X, Y \rangle|}{\|X\| \|Y\|} = \cos \theta$

MMSE: $\arg \min_{\hat{y}} E[\|X - \hat{y}\|^2] = E[X|Y]$ $\text{USE: } \hat{X}_{\text{USE}} = E[X] + \frac{\text{cov}(X, Y)}{\text{var}(Y)} (Y - E[Y])$

$\text{var}(Y) = \frac{E[Y^2] - (E[Y])^2}{P(Y)}$ $E[X|Y] = \int x p(x|y) dx \quad E[(X - L(X|Y))^2] = \text{var}(X - \frac{\text{cov}(X, Y)}{\text{var}(Y)} (Y - E[Y]))$

IF $(Y - \phi(X))^2$ minimize $Y - \phi(X)$ sum of X orthogonal USE update:

$E[Y - \phi(X)] = 0 \quad L[Y|X, Z] = L[Y|X] + L[Y|Z] \quad \hat{Z} = Z - L[Z|X]$

JOINT GAUSSIANS: MMSE = USE $\vec{z} = \text{standard normal vector}$

$x_1, \dots, x_n \text{ i.i.d. } A \text{ s.t. } \vec{x} = A \vec{z} + \mu$

$x_1, \dots, x_n \text{ i.i.d. } A \text{ s.t. } \vec{x} = A \vec{z} + \mu$ allows normal dist.

$\hat{x}(\vec{y}) = \frac{1}{\sqrt{(2\pi)^n \det(\Sigma)}} \exp\left(-\frac{1}{2}(\vec{y} - \mu)^T \Sigma^{-1}(\vec{y} - \mu)\right) \quad \Sigma = A A^T$

properties: Σ indep $\Leftrightarrow$ uncorrelated $E[X|Y] = \mu + \sum_i \Sigma_{ii}^{-1} (y_i - \mu_i)$

i.e. Σ cov Σ $\text{if } X \sim \mathcal{N}(\mu, \Sigma), A x + b \sim \mathcal{N}(A \mu + b, A \Sigma A^T)$

MMSE $E[X|Y] = \text{USE } L[X|Y] \quad \text{if } X \sim \mathcal{N}(\mu, \Sigma), A x + b \sim \mathcal{N}(A \mu + b, A \Sigma A^T)$

$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right) \quad \text{cov}(X|Y) = \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX}$

RANDOM PAX

$T_1 \sim \text{Exp}(\lambda_1) \quad T_2 \sim \text{Exp}(\lambda_2)$

$$P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

$$P(N(t) \text{ even}) = \frac{1 + e^{-2\lambda t}}{2} \rightarrow \frac{1}{2} \quad t \rightarrow \infty$$

$$\mathcal{N}(\mu, \sigma^2) = N + \sigma \mathcal{Z} \Rightarrow \frac{\mathcal{Z} - \mu}{\sigma}$$

MV δ :

$$L(\mu, \Sigma) = -\frac{1}{2} \log |2\pi \Sigma| - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i \quad \hat{\Sigma}_{MLE} = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})(x_i - \hat{\mu})^T$$

$$\text{rank}(\hat{\Sigma}_{MLE}) \leq n-1$$

normal-normal MAP

$$\hat{\mu}_{MAP} = \frac{k_0 \mu_0 + n \bar{x}}{k_0 + n} \quad \text{large } n \rightarrow \text{MAP} \approx \text{MLE}$$

normal-inverse-wishart prior $\text{large } k_0 \rightarrow \text{prior dom}$

$$\Sigma_n = k_0 \Sigma_0 + n \bar{S} \quad \hat{\Sigma}_{MAP} = \frac{\gamma_0 \Sigma_0}{\gamma_0 + n}$$

$$\mu_n = \frac{k_0 \mu_0 + n \bar{x}}{k_0 + n} \quad \hat{\Sigma}_{MAP} = \frac{\gamma_0 \Sigma_0}{\gamma_0 + n}$$

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$N(t)$ even: security guard back inside $N(t)$ odd: outside

$\frac{1}{2}$ chance he is already there $\frac{1}{2}$ guard is outside but returns by the time I leave

$\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{1+1}$

$$p(\text{no guards}) = \frac{1}{2^n} \quad \square$$

$$\Rightarrow \frac{N\lambda_1}{N\lambda_1 + \lambda_2} \Rightarrow \left(1 - \frac{1}{2^N}\right) + \frac{1}{2^N} \cdot \frac{N\lambda_1}{N\lambda_1 + \lambda_2}$$

break independent increments in $(N_t^{(i)})$ and $(N_t^{(j)}) \Rightarrow$ increments in original process independent $\&$ within each such interval, arrivals are thinned independently

verify independent increments: $(s_1, t_1], (s_2, t_2], \dots, (s_m, t_m]$

$$N_{tj} - N_{sj} = (N_{tj}^{(1)} - N_{sj}^{(1)}) + (N_{tj}^{(2)} - N_{sj}^{(2)})$$

independent to other intervals independent to other intervals

independent w.r.t. each other

$$p(x_2 > x_1) = E[\mathbb{1}_{x_1 > x_2}] = E[E[\mathbb{1}_{x_1 > x_2} | x_1]] = E[p(x_2, x_1 | x_1)] = E[e^{-\lambda_1 x_1}]$$

$$= \int_0^\infty e^{-\lambda_1 x_1} \lambda_1 e^{-\lambda_2 x_1} dx_1 = \lambda_1 \int_0^\infty e^{-(\lambda_1 + \lambda_2)x_1} dx_1 = \lambda_1 \cdot \frac{1}{\lambda_1 + \lambda_2} e^{-(\lambda_1 + \lambda_2)x_1} \Big|_0^\infty = \frac{\lambda_1}{\lambda_1 + \lambda_2} [e^0 - e^{-\infty}] = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

c) max = min + addtl. time for remaining cashier to finish
 $E[Y|Z] = z + p_1 E[X_2] + p_2 E[X_1] = z + \frac{p_1}{\lambda_2} + \frac{p_2}{\lambda_1}$

$$\begin{aligned} d) \text{ To find } \phi(x_1, x_2) &= \lambda_1 e^{-\lambda_1 x_1} \cdot \lambda_2 e^{-\lambda_2 x_2}, \quad x_1, x_2 \geq 0 \\ \phi(x_1, x_2) &= \int_0^{\infty} \int_0^{\infty} \mathbf{1}_{\{x_1 \leq x, x_2 \leq z\}} \phi(x_1, x_2) dx_1 dx_2 = \int_0^x \int_0^z \lambda_1 e^{-\lambda_1 x_1} \lambda_2 e^{-\lambda_2 x_2} dx_1 dx_2 \\ \phi(x_1, x_2) &= \frac{\partial}{\partial x} \frac{\partial}{\partial z} \int_0^x \int_0^z \lambda_1 e^{-\lambda_1 x_1} \lambda_2 e^{-\lambda_2 x_2} dx_1 dx_2 = \frac{\partial}{\partial x} \int_0^x \lambda_1 e^{-\lambda_1 x_1} \lambda_2 e^{-\lambda_2 (x-x_1)} dx_1 = \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 (x-x)} \end{aligned}$$

a) $\frac{1}{\binom{N}{k}} \Rightarrow N \geq \max_i X_i$
 $L(N|\vec{X}) = p(\vec{X}|N) = \frac{1}{\binom{N}{k}} \mathbb{1}_{\{N \geq x_{(k)}\}}$ where $x_{(k)} = \max_i X_i$
 $\binom{N}{k}$ is increasing for $N \geq k \Rightarrow$ maximized when N is smallest $\Rightarrow N = x_{(k)}$ $\hat{N}_{MLE} = x_{(k)}$

b) $\hat{N}_{MLE} = X_{(k)} = M$
 Need $E[N] \rightarrow E[M] : P(M=m) = \binom{m-1}{k-1} \cdot \frac{1}{M} \cdot \left(\frac{M-1}{M}\right)^{m-1}$ (max fixed, choose remaining $k-1$ from unobserving less than m)
 $E[M] = \sum_{m=k}^M m \cdot \binom{m-1}{k-1} \cdot \frac{1}{M} \cdot \left(\frac{M-1}{M}\right)^{m-1}$
 $\Rightarrow N = \frac{k-1}{k} E[M] \rightarrow \frac{k-1}{k+1} \cdot N = \frac{N \cdot k - N \cdot k}{k+1} = \frac{N \cdot k - N}{k+1} \Rightarrow N \cdot k \geq N$
 $\hat{N}_{MLE} = \frac{k+1}{k} M - 1$

$$y=0: \mathcal{L}(y=0) = \frac{p_{Y|X}(y=0|X=1)}{p_{Y|X}(y=0|X=0)} = \frac{\frac{1}{4}}{\frac{2}{4}} = \boxed{\frac{1}{3}} \quad y=1: \mathcal{L}(y=1) = \frac{p_{Y|X}(y=1|X=1)}{p_{Y|X}(y=1|X=0)} = \frac{\frac{3}{4}}{\frac{1}{4}} = \boxed{3}$$

For discrete RV:
choose λ to the
equal to likelihood
thresholds.

$\hat{\lambda}_p = \begin{cases} 0 & \mathbb{I}(y) < \frac{1}{3} \rightarrow \text{impossible (either 1 or 3)} \\ \text{Bern}(T) & \mathbb{I}(y) = \frac{1}{3} \\ 1 & \mathbb{I}(y) > \frac{1}{3} \rightarrow \text{must be 3 so choose 1} \\ 0 & \mathbb{I}(y) < \frac{1}{3} \rightarrow \text{must be 1 so choose 0} \end{cases}$

write down as many
decision rules as
thresholds

$\hat{X}_p = \begin{cases} \text{Bern}(x_2) & \mathcal{Z}(y) = 3 \\ \text{Bern}(x_2) & \mathcal{Z}(y) = 3 \end{cases}$

hint:

$\gamma P(L(y)=1|x=0) + P(L(y)>1|x=0) = \beta$
 Probability False Alarm $\rightarrow$ PFA: Accidentally picking 1 when $x=0$

Satz $\lambda = 3$: impossible! $= 0$

$$\beta_2 = \frac{\beta \cdot P(X(4)=3 | X=0)}{P(X(4)=3 | X=0)} \rightarrow X(4)=3 \Leftrightarrow y=1$$

$$\beta_2 \cdot P(y=1 | X=0) = \beta_2 \cdot \frac{1}{4} = \beta \quad \beta_2 = 4\beta \Rightarrow \beta \leq \frac{1}{4}$$

Satz $\lambda = \frac{3}{2}$: must be $X(4)=3$

$$\beta_1 = \frac{\beta \cdot P(X(4)=\frac{3}{2} | X=0)}{P(X(4)=\frac{3}{2} | X=0)} \rightarrow X(4)=\frac{3}{2} \Leftrightarrow y=0$$

$$= \frac{\beta \cdot P(X(4)=3 | X=0)}{P(y=0 | X=0)} = \frac{\beta \cdot \frac{1}{2}}{\frac{3}{4}} \quad \text{under null hypothesis}$$

$$\beta_1 \cdot \frac{3}{4} = \beta \cdot \frac{1}{2} \Rightarrow \beta = \frac{3}{2} \beta_1 \cdot \frac{1}{4} \quad 0 \leq \beta_1 \leq 1$$

so $\boxed{\beta \geq \frac{1}{4}}$

$$L[T|D] = E[T] + \frac{\text{cov}(T,D)}{\text{var}(D)^2} (D - E[D]) \quad D = T + N$$

$$\begin{aligned} \text{cov}(T, D) &= E[TD] - E[T]E[D] = E[T(T+N)] - E[T]E[T+N] \\ &= E[T^2] + E[TN] - E[T]^2 - E[T]E[N] \\ &= E[T^2] + E[T]E[N] - E[T]^2 - E[T]E[N] = E[T^2] - E[T]^2 = \text{var}(T) \end{aligned}$$

$$E[T] = p(\lambda) + (1-p)0 = \lambda p \quad T|ON \sim \text{Poisson}(\lambda)$$

$$\text{var}(T|ON) = \lambda \quad E[T|ON] = \lambda$$

$$\text{var}(T|O_N) = E[T^2|O_N] - E[T|O_N]^2 \Rightarrow E[T^2|O_N] = \lambda + \lambda^2$$

$$\mathbb{E}[\tau^2] = p(\lambda + \lambda^2) + (1-p)(0) = p(\lambda + \lambda^2)$$

$$\text{var}(T) = p(\lambda + \lambda^2) - (\lambda p)^2$$

$$\begin{aligned} \text{var}(D): \text{var}(T+N) &= \text{var}(T) + \text{var}(N) + 2\text{cov}(T, N) \\ &= \rho(\lambda + \lambda^2) - (\lambda\rho)^2 + \mu \end{aligned}$$

$$E[D] = E[T+N] = E[T] + E[N] = p\lambda + \mu$$

$$LLSE = p\lambda + \frac{p(\lambda + \lambda^2) - (\lambda p)^2}{(p(\lambda + \lambda^2) - (\lambda p)^2 + \mu^2)} (D - (p\lambda + \mu))$$

a) $g(Y) := L[X|Y]$ $g(Z) := L[X|Z]$

$X, f(Y), g(Z)$: zero mean

$$\langle u, v \rangle = \text{tr}[uv]^Y \perp \mathfrak{g}(z) \in \mathfrak{sp}(1, z) \quad z \perp \mathfrak{h}(Y) \in \mathfrak{sp}(1, Y)$$

$$E[Y \cdot 1] = E[Y] = 0$$

$$\mathbb{E}[x - f(y) - g(z)] = 0$$

$$E[(x - f(y) - g(z))y] = 0 \text{ (residual after removing } x \text{ projected onto } y)$$

$$\mathbb{E}[(x - f(y) - g(z))z] = 0$$

$x - f(y) - g(z)$ is orthogonal to any linear function of y , z , and x :

$$L[x|y, z] = f(y) + g(z)$$

$$= L[X|Y] + L[X|Z]$$

b) $z - L[z|Y] \perp Y$

Let $w := z - L[z|Y]$

$$\Rightarrow L(x|y, w) = L(x|y) + L(x|w)$$

$$\Rightarrow L[X|Y, W] = L[X|Y, Z] \Rightarrow \text{sp}\{1, Y, W\} = \text{sp}\{1, Y, Z\}$$

$$\therefore L[X|Y, Z] = L[X|Y] + L[X|(Z - L[Z|Y])]$$